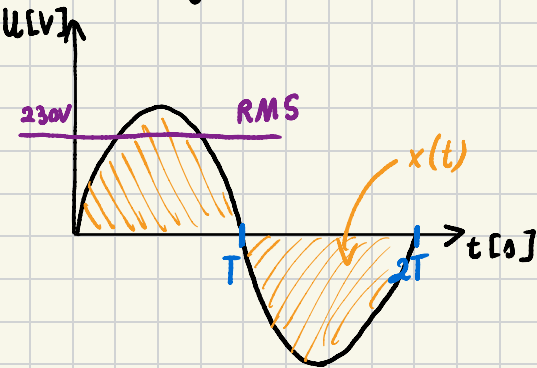


ĆWICZENIA 1

Znl: kilka pytań, jedno kolokwium

Przebieg prądu

Europa → 230V, 50Hz

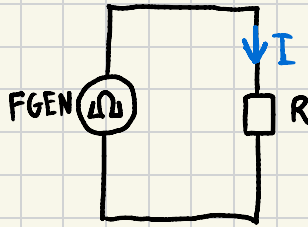
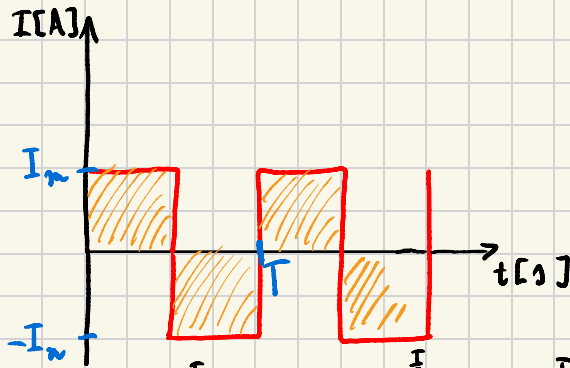


$$x_{SR} = \frac{1}{T} \int_0^T x(t) dt$$

$$x_{RMS} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}$$

NAP. SKUTECZNE

Zad 1

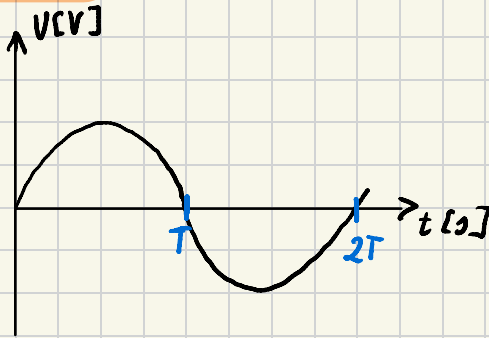


$$\begin{aligned} x_{SR} &= \frac{1}{T} \int_0^T x(t) dt = \frac{1}{T} \int_0^{\frac{T}{2}} I_m + \frac{1}{T} \int_{\frac{T}{2}}^T -I_m dt = \frac{1}{T} [I_m \cdot t]_0^{\frac{T}{2}} - \frac{1}{T} [I_m \cdot t]_{\frac{T}{2}}^T = \\ &= \frac{1}{T} (I_m \cdot \frac{T}{2} - 0) - \frac{1}{T} (I_m \cdot T - I_m \cdot \frac{T}{2}) = \frac{1}{2} I_m - I_m + \frac{1}{2} I_m = 0 \end{aligned}$$

Przebiegi na zaliczeniu

↳ mogą jakieś być, po prostu policz całkę

Zad 2

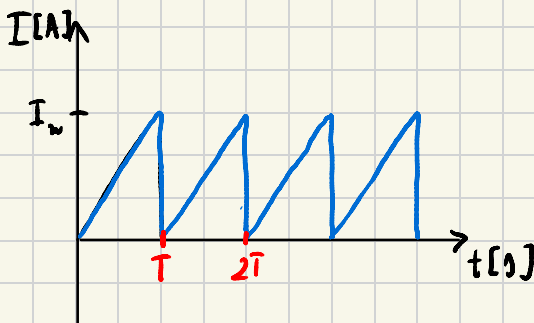


$$x(t) = \sin(\omega t)$$

$$x_{GR} = \frac{1}{T} \int_0^T x(t) dt = \frac{1}{T} \int_0^T \sin(\omega t) dt = -\frac{1}{\omega T} [\cos(\omega t)]_0^T =$$

$$= -\frac{1}{\omega T} [\cos(\omega T) - 1] = \frac{1}{\omega T} [1 - \cos(\omega T)]$$

Zad 3



$$I_{RMS} = \sqrt{\frac{1}{T} \int_0^T I^2(t) dt} = \sqrt{\frac{1}{T} \left[\frac{I_m^2}{T^2} \cdot \frac{t^3}{3} \right]_0^T} =$$

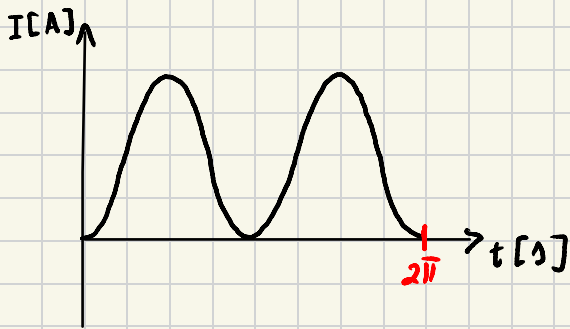
$$I(t) = a \cdot t \quad = \frac{I_m}{T}$$

$$I(T) = I_m = a \cdot T \cdot a = \frac{I_m}{T}$$

$$I(t) = \frac{I_m}{T} \cdot t$$

$$\int \left(\frac{I_m}{T} \cdot t \right)^2 dt = \left(\frac{I_m}{T} \right)^2 \int t^2 dt = \left(\frac{I_m}{T} \right)^2 \cdot \frac{t^3}{3}$$

Zad. 4



$$x(t) = I_m \cdot \sin \omega t$$

$$x_q(t) = x^2(t)$$

$$I_{RMS} = \sqrt{\frac{1}{T} \int_0^T I_m^2 \cdot \sin^2 \omega t \, dt} = \sqrt{\frac{I_m^2}{T} \int_0^T \sin^2 \omega t \, dt} =$$

$$\int_0^T \sin^2 \omega t \, dt =$$